

$$\begin{bmatrix} 3 & 2 & 1 & -5 & 0 \\ -9 & -4 & 1 & 7 & 0 \\ 9 & 2 & -5 & 1 & 0 \end{bmatrix} \xrightarrow[\text{3} \times \text{Row 1}]{\text{Row 2} = \text{Row 2} +} \begin{bmatrix} 3 & 2 & 1 & -5 & 0 \\ 0 & 2 & 4 & -8 & 0 \\ 9 & 2 & -5 & 1 & 0 \end{bmatrix} \xrightarrow[\text{3} \times \text{Row 1}]{\text{Row 3} = \text{Row 3} -}$$

$$\begin{bmatrix} 3 & 2 & 1 & -5 & 0 \\ 0 & 2 & 4 & -8 & 0 \\ 0 & -4 & -8 & 16 & 0 \end{bmatrix} \xrightarrow[\times 1/2]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 3 & 2 & 1 & -5 & 0 \\ 0 & 1 & 2 & -4 & 0 \\ 0 & -4 & -8 & 16 & 0 \end{bmatrix} \xrightarrow[\times (-1/4)]{\text{Row 3} = \text{Row 3} +}$$

$$\begin{bmatrix} 3 & 2 & 1 & -5 & 0 \\ 0 & 1 & 2 & -4 & 0 \\ 0 & 1 & 2 & -4 & 0 \end{bmatrix} \xrightarrow[\text{-Row 2}]{\text{Row 3} = \text{Row 3} -} \begin{bmatrix} 3 & 2 & 1 & -5 & 0 \\ 0 & 1 & 2 & -4 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{-2} \times \text{Row 2}]{\text{Row 1} = \text{Row 1} -}$$

$$\begin{bmatrix} 3 & 0 & -3 & 3 & 0 \\ 0 & 1 & 2 & -4 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[\times 1/3]{\text{Row 1} = \text{Row 1}} \begin{bmatrix} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 2 & -4 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The final matrix is in rref. It's easy to see that only column 1 and 2 are pivot columns, $\therefore$ by defn, x_3 and x_4 are free variables.

So, the corresponding system of equations are: $x_1 - x_3 + x_4 = 0$ ①

② $x_2 + 2x_3 - 4x_4 = 0$ ③ $0 = 0$

$\therefore x_1 = x_3 - x_4$ $x_2 = 4x_4 - 2x_3$. Let, $x_3 = x_4 = 1$ $\therefore x_1 = 1 - 1 = 0$
 $x_2 = 4 \cdot 1 - 2 \cdot 1 = 2$.

$x = \begin{bmatrix} 0 \\ 2 \\ 1 \\ 1 \end{bmatrix} \in \text{Nul } A \quad \square$

Determine which sets in Exercises 15-20 are bases for $\mathbb{R}^2$ or $\mathbb{R}^3$. Justify each answer.

15. $\begin{bmatrix} 5 \\ -2 \end{bmatrix}, \begin{bmatrix} 10 \\ -3 \end{bmatrix}$ Solution: Let $A = \begin{bmatrix} 5 & 10 \\ -2 & -3 \end{bmatrix}$ $\det A = 5(-3) - 10(-2) = -15 + 20 = 5 \neq 0$
 $\therefore$ From Theorem 4, $A^{-1} = \frac{1}{5} \begin{bmatrix} -3 & -10 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} -3/5 & -2 \\ 2/5 & 1 \end{bmatrix}$

By the Invertible Matrix Theorem and the equivalence of the statements (a), (e) and (h), the columns of A form a linearly independent set and they span $\mathbb{R}^2$. $\square$